

Q1 2021 (31 Aug Shift 1)

A wire having a linear mass density $9.0 \times 10^{-4} \text{ kg/m}$ is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is m.

Q2 2021 (27 Aug Shift 2)

A tuning fork is vibrating at 250 Hz. The length of the shortest closed organ pipe that will resonate with the tuning fork will be cm. (Take speed of sound in air as 340 ms^{-1})

Q3 2021 (27 Aug Shift 1)

Two cars X and Y are approaching each other with velocities 36 km/h and 72 km/h respectively. The frequency of a whistle sound as emitted by a passenger in car X, heard by the passenger in car Y is 1320 Hz. If the velocity of sound in air is 340 m/s, the actual frequency of the whistle sound produced is Hz

Q4 2021 (26 Aug Shift 2)

Two waves are simultaneously passing through a string and their equations are :

$y_1 = A_1 \sin k(x - vt)$, $y_2 = A_2 \sin k(x - vt + x_0)$. Given amplitudes $A_1 = 12 \text{ mm}$ and $A_2 = 5 \text{ mm}$

$x_0 = 3.5 \text{ cm}$ and wave number $k = 6.28 \text{ cm}^{-1}$. The amplitude of resulting wave will be mm

Q5 2021 (26 Aug Shift 1)

Two travelling waves produces a standing wave represented by equation,

$$y = 1.0 \text{ mm} \cos(1.57 \text{ cm}^{-1}) \times \sin(78.5 \text{ s}^{-1})t$$

The node closest to the origin in the region $x > 0$ will be at $x = \dots\dots\dots \text{cm}$.

Q6 2021 (26 Aug Shift 1)

A source and a detector move away from each other in absence of wind with a speed of 20 m/s with respect to the ground. If the detector detects a frequency of 1800 Hz of the sound coming from the source, then the original frequency of source considering speed of sound in air 340 m/s will be Hz

Answer Key

Q1 (10)

Q2 (34)

Q3 (1210)

Q4 (7)

Q5 (1)

Q6 (2025)

Q1 (10)

$$\mu = 9.0 \times 10^{-4} \frac{\text{kg}}{\text{m}}$$

$$T = 900 \text{ N}$$

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{900}{9 \times 10^{-4}}} = 1000 \text{ m/s}$$

$$f_1 = 500 \text{ Hz}$$

$$f = 550$$

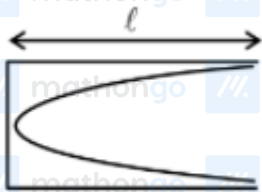
$$\frac{nV}{2\ell} = 500 \dots (i)$$

$$\frac{(n+1)V}{2\ell} = 500 \dots$$

$$(ii) (i) \frac{V}{2\ell} = 50$$

$$\ell = \frac{1000}{2 \times 50} = 10$$

Q2 (34)



$$\frac{\lambda}{4} = \ell \Rightarrow \lambda = 4\ell$$

$$f = \frac{v}{\lambda} = \frac{v}{4\ell}$$

$$\Rightarrow 250 = \frac{340}{4\ell}$$

$$\Rightarrow \ell = \frac{34}{4 \times 25} = 0.34 \text{ m}$$

$$\ell = 34 \text{ cm}$$

Q3 (1210)



$$V_x = 36 \text{ km/hr} = 10 \text{ m/s}$$

$$V_y = 72 \text{ km/hr} = 20 \text{ m/s}$$

by doppler's effect

$$F' = F_0 \left(\frac{V \pm V_0}{V \pm V_s} \right)$$

$$1320 = F_0 \left(\frac{340+20}{340-10} \right) \Rightarrow F_0 = 1210 \text{ Hz}$$

Q4 (7)

$$y_1 = A_1 \sin(x - vt)$$

$$y_1 = 12 \sin 6.28(x - vt)$$

$$y_2 = 5 \sin 6.28(x - vt + 3.5)$$

$$\Delta\phi = \frac{2\pi}{\lambda}(\Delta x)$$

$$= K(\Delta x)$$

$$= 6.28 \times 3.5 = \frac{7}{2} \times 2\pi = 7\pi$$

$$A_{\text{nd}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$A_{\text{net}} = \sqrt{(12)^2 + (5)^2 + 2(12)(5) \cos(7\pi)}$$

$$= \sqrt{144 + 25 - 120}$$

Ans. (7)

Q5 (1)

For node

$$\cos(1.57 \text{ cm}^{-1})x = 0$$

$$(1.57 \text{ cm}^{-1})x = \frac{\pi}{2}$$

$$x = \frac{\pi}{2(1.57)} \text{ cm} = 1 \text{ cm}$$

Ans. 1.00

Q6 (2025)



$$f' = f \left(\frac{C - V_O}{C + V_S} \right)$$

$$1800 = f \left(\frac{340 - 20}{340 + 20} \right)$$

$$f = 2025 \text{ Hz}$$

Ans. 2025